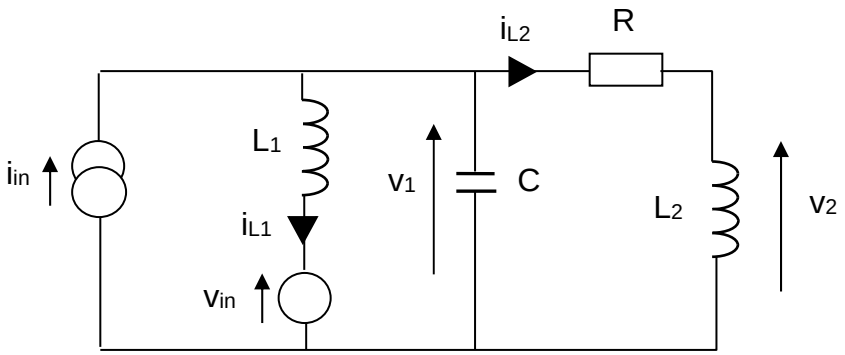
Problem 1.1 (Theoretical and MATLAB)
State space model and transfer function for continuous-time LTI system

Consider the following passive electrical circuit (continuous-time LTI system) with 2 inputs and 1 output



- Find the transfer functions $H_1(s) = \frac{V_2(s)}{I_{in}(s)}$ and $H_2(s) = \frac{V_2(s)}{V_{in}(s)}$ using s-domain impedances and the rules for parallel/serial connection and voltage/current division

---

Next, it is desired to derive a continuous-time state space model (SSC1) for the circuit on the following form

$$\frac{d\mathbf{x}}{dt} = A\,\mathbf{x}(t) + B\,\mathbf{u}(t)\,, \qquad y(t) = C\,\mathbf{x}(t) + D\,\mathbf{u}(t)$$

where - state vector $\mathbf{x}(t) = \begin{bmatrix} i_{L1}(t) \\ v_1(t) \\ i_{L2}(t) \end{bmatrix}$

- input vector $\mathbf{u}(t) = \begin{bmatrix} i_{in}(t) \\ v_{in}(t) \end{bmatrix}$

- output $y(t) = v_2(t)$

- Indicate the dimensions of matrices A, B, C, D
- Find the SSC1 model, that is, find the matrices A, B, C, D (Hint: for each of the 3 reactive elements in the circuit, state the time-domain component equation, e.g. v = L*di/dt and i = C*dv/dt, put the derivatives on the left hand side and express them as linear combinations of states and inputs)

The following element values are now defined: $R = 1k\Omega$
$L_1 = 1mH$
$L_2 = 2mH$
$C = 1\mu F$

- Define in MATLAB the transfer functions $H_1(s)$ and $H_2(s)$ – either as nominator and denominator coefficient vectors (of same length) and/or by creating transfer function objects
- Find the transfer function poles

- Define in MATLAB the SSC1 model – either by just defining the matrices A, B, C, D and/or by creating a state space object
- Find the eigenvalues for the state transition matrix A and compare with transfer function poles
- Verify numerically for the eigenvalues λ_i: $\sum_i \lambda_i = \mathrm{trace}(A)$
$\prod_i \lambda_i = |A|$

Next, for transformation between LTI system representations

- Transform the SSC1 model to transfer functions – either by manipulating matrices and coefficient vectors directly (***ss2tf***) and/or by converting the state space object to transfer function objects (***tf***)
- Compare the resulting transfer functions with the previously derived and defined transfer functions

- For the opposite transformation, find an alternative state space model (SSC2) for the circuit by transforming the transfer functions $H_1(s)$ and $H_2(s)$ to a new state space object (***ss***)
- Compare the matrices for SSC1 and SSC2
- Find the eigenvalues for the SSC2 A-matrix and compare with the eigenvalues for the SSC1 A-matrix

---

As a little extra teaser consider the ideal continuous-time integrator and differentiator with transfer functions $H(s) = 1/s$ and $H(s) = s$, respectively.

- For the integrator and the differentiator, define transfer function objects (***tf***), convert to state space models (***ss***)
- Explain (with equations) why the state space equations with the found matrices implement an integrator and differentiator, respectively (a little harder)
- How can the contents of the SSC state(s) be interpreted/explained ?

Problem 1.2 (Theoretical and MATLAB)
State space model and transfer function for discrete-time LTI system

A discrete time LTI system (e.g. digital filter) with input signal u(n) and output signal y(n) is defined by the difference equation

$$y(n) = b_0u(n) + b_1u(n-1) + b_2u(n-2) + b_3u(n-3) - a_1y(n-1) - a_2y(n-2) - a_3y(n-3)$$

The coefficients are:

$b_0 = 1$

$b_1 = 2$ $\quad a_1 = -3/4$

$b_2 = -2$ $\quad a_2 = -1/4$

$b_3 = 1$ $\quad a_3 = 3/16$

- Write (on paper) the transfer function $H(z) = \frac{Y(z)}{U(z)}$
- Define in MATLAB the transfer function H(z) – either as nominator and denominator coefficient vectors (of same length) and/or by creating a transfer function object using ***tf*** (make sure that the defined object is discrete-time by indicating an (unspecified) sample period (Ts = [ ]) and setting the 'variable' property to either 'z' or 'z^{-1}')
- Find the transfer function poles

Based on the transfer function H(z), it is desired to derive a discrete-time state space model (SSD1) for the LTI system on the following form

$$\mathbf{x}(n+1) = A\,\mathbf{x}(n) + B\,u(n)\,, \qquad y(n) = C\,\mathbf{x}(n) + D\,u(n)$$

- Indicate the dimensions of matrices A, B, C, D
- Derive from H(z) (on paper) a SSD1 model for the LTI system, (find A,B,C,D)
  (Hint: follow optionally this procedure: first introduce a dummy variable v(n) by
  H(z) = Y(z)/U(z) = [z-nominator / z-denominator] * [V(z) / V(z)],
  next equate left and right hand side nominators and denominators,
  inverse z-transform both equations,
  choose state vector $\mathbf{x}(n) = [v(n)\ v(n+1)\ v(n+2)]^T$ and finally
  write $\mathbf{x}(n+1)$ and y(n) as linear combinations of states and input)
- Define in MATLAB the SSD1 model – either by just defining the matrices A, B, C, D and/or by creating a state space object using ***ss*** (make sure that the defined object is discrete-time by indicating an (unspecified) sample period (Ts = [ ])
- Find the eigenvalues for the state transition matrix A and compare with transfer function poles
- Verify numerically for the eigenvalues λ_i: $\sum_i \lambda_i = \mathrm{trace}(A)$, $\prod_i \lambda_i = |A|$

Next, for transformation between LTI system representations

- Transform the SSD1 model to a transfer function – either by manipulating matrices and coefficient vectors directly (***ss2tf***) and/or by converting the state space object to a transfer function object (***tf***)
- Compare the resulting transfer function with the previously defined transfer function

- For the opposite transformation, find an alternative state space model (SSD2) for the system - either by manipulating matrices and coefficient vectors directly (***tf2ss***) and/or by transforming the transfer function H(z) to a new state space object (***ss***)
- Compare the matrices for SSD1 and SSD2
- Find the eigenvalues for the SSD2 A-matrix and compare with the eigenvalues for the SSD1 A-matrix

---

As yet another alternative state space model for the system, it is desired to find the modal representation, that is, a state space model where the A-matrix is diagonal and contains the A-eigenvalues.

- Thus, define a state transformation from state vector **x**(n) to a new state vector **z**(n) by the linear transformation E **z**(n) = **x**(n), where the columns of the transformation matrix E are the eigenvectors of A
- Find the A_z, B_z, C_z, D_z matrices for the state space model with state vector **z**(n)

As a little extra teaser consider the first-order approximations to a discrete-time integrator and differentiator with transfer functions $\mathrm{H}(z) = 1/(1-\mathrm{z}^{-1})$ and $\mathrm{H}(z) = 1-\mathrm{z}^{-1}$, respectively.

- For the integrator and the differentiator, define transfer function objects with ***tf*** (again, make sure that the defined objects are discrete-time by indicating an (unspecified) sample period (Ts = [ ]) and setting the 'variable' property to either 'z' or 'z^{-1}')
- Convert to state space models (***ss***)
- Explain (with equations) why the state space equations with the found matrices implement (approximations to) an integrator and differentiator, respectively
- How can the contents of the SSD state(s) be interpreted/explained ?
